	Parul University Faculty of Engineering and Technology Parul Institute of Engineering and Technology Department of Artificial Intelligence and Data Science			
Subject Name	**Theory of Computations**	**A.Y**	**2026-27**	
Subject Code	**303105306**	**Semester**	**5**	

Assignment-5

Sr No	Question	COs	B.T	Competence
1	Define the Church–Turing Thesis and explain its significance in the Theory of Computation.	1	1	Remember
2	Explain the concept of a Universal Turing Machine (UTM) with suitable examples. Describe how a UTM simulates another Turing Machine.	4	2	Understand
3	Illustrate the working of a Universal Turing Machine for a simple input and explain the role of encoding in TM simulation.	4	3	Apply
4	Analyze the Universal Language and Diagonalization Language with respect to decidability and computability.	5	4	Analyze
5	Evaluate how the diagonalization method is used to prove undecidable problems in computation theory. Justify your answer with suitable reasoning.	5	5	Evaluate
6	Design a flow representation or model showing the relationship among Church–Turing Thesis, Universal Turing Machine, Universal Language, and Diagonalization Language in the study of undecidability.	5	6	Create